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Flow Control at DataLink Layer using Sliding window protocol

Aim:

To write a program to implement flow control at the datalink Layer using the sliding window protocol

Algorithm:

1. Start the program
2. Read the total number of frames to be sent(N) the Senders window size (w) and (optionally) a frame number to simulate as lost during transmission.
3. Initialize base = 0 (Sequence number of the oldest unacknowledged frame) and next seq = 0 (Sequence no. of the next frame to be sent)
4. Repeat while base < N:
   a.) Determine the current sender window as the range [base, base + w - 1].
   b.) Send all frames with sequence numbers from next seq up to (base + w - 1), incrementing next seq after each frame, since the window restricts the Sender to at most w outstanding (unacknowledged) frames at anytime.

OUTPUT

```
=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (Go-Back-N) ===

Enter total number of frames to send     : 8
Enter sender's window size               : 4
Enter frame number to simulate as LOST
(0 to 7, enter -1 for no loss)           : 5

-----------------------------------------------
Sender Window : [Frame 0 - Frame 3]
Node A -->Sending Frame 0
Node A -->Sending Frame 1
Node A -->Sending Frame 2
Node A -->Sending Frame 3
Node B -->Received Frame 0, sending ACK 0
Node B -->Received Frame 1, sending ACK 1
Node B -->Received Frame 2, sending ACK 2
Node B -->Received Frame 3, sending ACK 3
Node A -->ACK 3 received. Sliding window forward.

-----------------------------------------------
Sender Window : [Frame 4 - Frame 7]
Node A -->Sending Frame 4
Node A -->Sending Frame 5
        (Frame 5 LOST in transit!)
Node A -->Sending Frame 6
Node A -->Sending Frame 7
Node B -->Frame 5 missing. Discarding out-of-order frames after it.
Node A -->Timeout! Resending window starting from Frame 5.

-----------------------------------------------
Sender Window : [Frame 4 - Frame 7]
Node A -->Sending Frame 5
Node A -->Sending Frame 6
Node A -->Sending Frame 7
Node B -->Received Frame 4, sending ACK 4
Node B -->Received Frame 5, sending ACK 5
Node B -->Received Frame 6, sending ACK 6
```



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Node B -->Received Frame 7, sending ACK 7
Node A -->ACK 7 received. Sliding window forward.

-----------------------------------------------
All 8 frames successfully transmitted from Node A to Node B.
```

c.) If the lost frame lies within the frames just sent:

(i) The receiver discards the lost Frame and any subsequent out of order frame.

(ii) The sender times out and retransmits all frames starting from the lost frame (Go-Back-N)

d.) Else (no loss in this batch):

(i) The receiver accepts each frame in order and sends a cumulative acknowledgement (Ack) for it.

(ii) on receiving the Ack for the last frame sent, the sender slides its window forward: base = next_seq.

5. Repeat step 4 until all N frames have been transmitted and acknowledged.

6.) Stop the program.

Program:

```
def Simulate(total_frames, window_size, lost_frame):
    base=0
    next_seq=0
    while base < total_frames:
        window_end = min(base + window_size - 1, total_frames
        print("\n----------")
        print(f"Sender window: [Frame {base} - Frame {
                                                    -end}]
        while next_seq < base + window_size
```

```
print(f"Node A -> Sending Frame {next_seq}")
if next_seq == lost_frame:
    print(f"  (Frame {next_seq} Lost In transit !)")
next_seq += 1
if lost_frame is not None and base <= lost_frame
< next_seq:
    print(f"Node B -> Frame {lost_frame} missing. Discarding
        out-of order frames after it")
    print(f"Node A -> Time out ! Resending window starting from
        Frame {lost_frame}")
next_seq = lost_frame
lost_frame = None
for f in range(base, next_seq):
print(f"Node B -> Received Frame {f}. Sending Ack {f}")
Print{f"Node A -> Ack {next_seq-1} received, sliding
        window forward"}
base = next_seq
Print("In ..... ")
print(f"All {total_frame} frame successfully transmitted
from Node A to Node B.")

def main():
    print("=== Sliding window protocol - Flow control
        simulation (Go Back-N) ===")
    total_frames = int(input("Enter total no. of frames to
        send:"))
window_size = int(input("Enter Senders window size
lost_frame = int(input(f"Enter frame number to simulate
    as Lost \n(0 to {total_frame-1}.
```

entry -1 for no loss):

If loss_frame == -1

lost_frame = none

Result:

The program to implement flow control at the Data link layer using the Sliding window protocol was successfully written in python, executed and verified by simulating the flow of frames